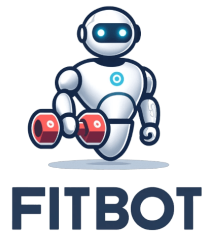


# Deep Learning

FitBot



MEI - 2023/2024

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# Contents

|          |                                           |           |
|----------|-------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                       | <b>2</b>  |
| <b>2</b> | <b>Domain Analysis and Objectives</b>     | <b>2</b>  |
| 2.1      | Domain Analysis . . . . .                 | 2         |
| 2.2      | Objectives . . . . .                      | 3         |
| <b>3</b> | <b>Project Structure</b>                  | <b>4</b>  |
| <b>4</b> | <b>Architecture</b>                       | <b>5</b>  |
| <b>5</b> | <b>Data Sources</b>                       | <b>7</b>  |
| 5.1      | Data Selection . . . . .                  | 7         |
| 5.2      | Data Processing . . . . .                 | 9         |
| <b>6</b> | <b>Results and Critical Analysis</b>      | <b>9</b>  |
| <b>7</b> | <b>Conclusion and Future Improvements</b> | <b>11</b> |

# 1 Introduction

This report serves as part of the practical project for the **Deep Learning** course within the Master's in Computer Engineering program at the University of Minho.

The evolution of Natural Language Processing (NLP) and Machine Learning (ML) technologies has opened new possibilities for creating more intelligent and useful virtual assistants. In this context, Large Language Models (LLMs) like Llama2, developed by Ollama, have proven to be particularly effective in understanding and generating text in a very natural and accurate way. At the same time, **Retrieval-Augmented Generation** (RAG) has emerged as a promising technique to enhance the performance of these models by integrating external and domain-specific information.

This practical project focuses on integrating RAG technology with LLMs to develop an advanced chatbot. The goal is to create a virtual assistant capable of crafting personalized training and nutrition plans, based on the scientific research that has been done over the years on these particular subjects, as well as providing useful tips on food and physical exercises. This solution aims not only to increase accessibility to specialized information but also to promote a healthier and well-informed lifestyle.

The RAG methodology allows the chatbot to access a vast amount of specific data on training and nutrition, using this information to generate more accurate and relevant responses. On the other hand, the use of LLMs will enable natural language interaction with users, ensuring an intuitive and efficient user experience.

In this work, the following steps will be addressed:

1. **Research and Data Selection:** Identification and organization of reliable data sources on nutrition and physical exercises.
2. **Development of the RAG-LLM Model:** Configuration and training of the model by integrating the capabilities of RAG with natural language generation of LLM.
3. **Chatbot Development:** Implementation of the chatbot using the developed model, focusing on providing precise and personalized responses.
4. **Testing and Validation:** Evaluation of the chatbot's performance through practical tests and user feedback.
5. **Graphical User Interface Development:** Implementation of a User Interface that allows the user to interact with the chatbot in a Web Based Application

## 2 Domain Analysis and Objectives

In this section, we will focus on describing the objectives of this project, more specifically the objectives of the chatbot, as well as analyse the research domain in which this chatbot will assist the users.

### 2.1 Domain Analysis

To create a chatbot capable of crafting workout and nutrition plans, as well as providing tips on food and exercises, it is essential to conduct in-depth research in the following areas:

#### **Physical Workout:**

1. **Science-Based Exercises:** Oversee detailed research on effective exercises grounded in scientific evidence, aiming to replace "old school" training practices with more modern and proven methods.

2. **Training Routines:** Analyze best practices for creating personalized training routines, considering variables such as age, fitness level, personal goals and physical limitations.
3. **Scientific Evidence:** Review scientific studies that support the effectiveness of different training methods.
4. **Types of Workout:** Research the various types and goals of workout, like *Hypertrophy Training* (goal of increase muscle size), *Strength Training* (goal of increase muscle strength) and *Endurance Training* (goal of improve cardiovascular and muscular endurance).

#### **Nutrition:**

1. **Macro and Micronutrients:** Understand the role of macronutrients (proteins, carbohydrates, and fats) and micronutrients (vitamins and minerals) in the diet.
2. **Meal Planning:** Methods for creating balanced and personalized meal plans, adjusted to individuals' energy and nutritional needs, based on recent studies.
3. **Specific Diets:** Investigate different types of diets (e.g., vegetarian, vegan, ketogenic) and their health implications.

#### **Chatbot and NLP Technology:**

1. **Large Language Models (LLMs):** Study the capabilities and limitations of the Ollama family LLMs, which can be altered by the user, in generating natural language text.
2. **Retrieval-Augmented Generation (RAG):** Explore RAG techniques to integrate external and domain-specific information into LLMs.
3. **Human-Computer Interaction:** Analyze best practices for creating intuitive and effective user interfaces for chatbots.

## **2.2 Objectives**

As mentioned before, the main objective of this project is to develop an advanced chatbot that utilizes the integration of RAG with LLMs to provide personalized training and nutrition recommendations. To achieve this goal, the following specific objectives have been defined:

#### **Data Identification and Organization:**

1. **Collect and Structure Data:** Gather and organize a comprehensive and reliable database on physical exercises and nutrition, using files and websites with scientifically grounded information.
2. **Ensure Scientific Accuracy:** Ensure that the data used is based on scientific evidence and is up-to-date.

#### **Development of the RAG-LLM Model:**

1. **Configure the Model:** Set up a model that combines RAG techniques with the natural language generation capabilities of the Ollama family LLMs.
2. **Train the Model:** Train the model with domain-specific data to improve the accuracy and relevance of responses, allowing for user modifications as necessary.

#### **Chatbot Implementation:**

1. **Develop User Interface:** Create a user-friendly and intuitive graphical interface for the chatbot.

2. **Deploy the Chatbot:** Deploy the chatbot on the web to make it accessible to users.
3. **Integrate the Model:** Integrate the RAG-LLM model into the chatbot, enabling it to provide personalized and data-driven recommendations.

#### Testing and Evaluation:

1. **Conduct Practical Tests:** Perform practical tests to evaluate the chatbot's effectiveness in different scenarios and with different user profiles.
2. **Analyze Performance:** Analyze the response time and accuracy of the chatbot's answers.

In conclusion, by integrating RAG technology with Ollama family LLMs, this project aims to create a chatbot that not only answers questions but also acts as a true personal assistant for training and nutrition. Through robust domain research and clear objectives, the developed system is expected to offer real value to users, promoting a healthier and more well-informed lifestyle.

## 3 Project Structure

This project is organized into six primary directories: **src**, **PDF Files**, **Final PDF Files**, **TXT Files**, **Embeddings** and **Images**. The **src** directory contains the Source Code of the Project. The **PDF Files** directory contains the initial PDF files, before the cleaning process. The **Final PDF Files** directory contains the PDF files, converted from the txt files, that are loaded to the Database to be used by the LLM Models. The **TXT Files** directory contains the txt files that were extracted from the PDF Files. The **Embeddings** directory contains the vector embeddings generated by the LLM Models and the Images directory contains the images used to build the Web Application of the model.

Additionally, the **Embeddings** directory also contains a folder for each model that are being used in this project, so that the evaluation process can occur in the most correct way. So, as we are using three models, *mistral*, *llama2* and *zephyr*, this directory contains the directories **Embeddings\_mistral**, **Embeddings\_llama** and **Embeddings\_zephyr**.

In the **src** directory we can find:

1. **app.py:** Contains the core chatbot implementation using Ollama LLM's.
2. **document\_loader.py:** Contains the script for loading the PDF files to the database.
3. **evaluate.csv:** Contains the Test Dataset for the evaluation of the LLM Models.
4. **evaluate.ipynb:** Contains the Jupyter Notebook for evaluating the LLM Models.
5. **extract\_pdfplumber.py:** Contains the code for extracting raw text from PDF files using PDFPlumber.
6. **streamlit.py:** Main script to run the chatbot on a Web Based Application.
7. **llm.py:** Contains the prompts and the chain build with the RAG Framework.
8. **models.py:** Contains code that verifies and downloads the LLM Models if not present in the local machine.
9. **py2pdf.py:** Contains the code for extracting raw text from PDF files using Py2PDF.
10. **requirements.txt:** Contains the tools necessary to run the project.
11. **Stats.csv:** Contains the Dataset that aggregates the ratings of the models achieved in **evaluate.ipynb**.

12. **stats.py**: Contains the script for visualizing the performance of the LLM models in a Streamlit Web Application.
13. **tesseract.py**: Contains the code for extracting raw text from PDF files using Tesseract.
14. **txt\_converter.py**: Contains the code that applies lowercase to the txt file provided.
15. **txt\_to\_pdf.py**: Contains the code that transforms a txt file to a PDF file.

## 4 Architecture

In this section, a description of the system will be presented, accompanied by diagrams to provide a better understanding of its architecture and operation. Throughout this section, the interactions between the Ollama LLM models, the ChromaDB database, the Langchain tool and the Graphical User Interface will be analysed, for a better understanding of the system.

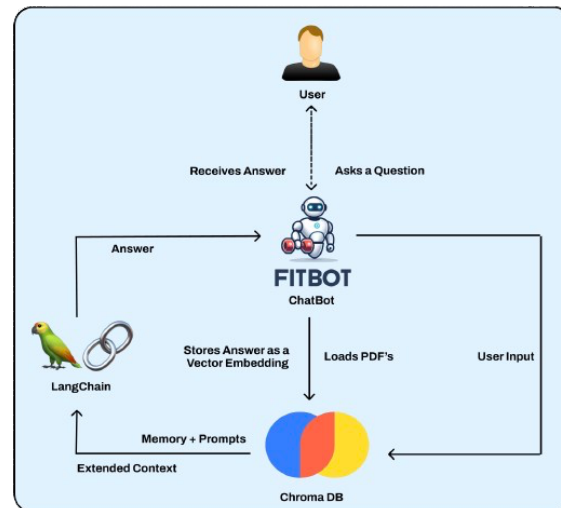


Figure 1: System Architecture

As we can see in the Figure 1, the Architecture of the system is composed by three main components: the **Chatbot Interface** (implemented with the Streamlit library), the **Database** (which is ChromaDB) and **LangChain Framework** (which makes use of the Ollama LLM Models). Each one of this components is very important for the structure of the system and have very distinct functions, as such:

1. **ChatBot Interface**: Interacts with the user as well as with ChromaDB and LangChain and is responsible for receiving user questions, send the user inputs to the database and deliver the answer to the User.
2. **ChromaDB**: Stores and manages all the data needed for the ChatBot, including PDFs, questions and answers that were previously generated by the LLM model, storing them in the form of vector embeddings. It also acts as an intermediary between FitBot and LangChain, facilitating data storage and retrieval.
3. **LangChain**: A language model framework that processes user input, extended context generated by RAG, and memory to generate accurate answers. Utilizes advanced language processing capabilities to understand and respond to user inputs.

4. **Ollama LLM Models:** These models are used by LangChain to generate the answer to the user's question, by feeding them with the extra content selected by the retriever in the Retrieval process, the memory which comes from the previous generated answers and the user question.

These previously mentioned tools were not chosen at random; in other words, there are several reasons why they were selected, which are as follows:

**ChromaDB:** It is a database made for easy and manageable access to vector embeddings, which were chosen as the primary way to store the data fed to the LLM Models by a couple of reasons, which are:

1. **Efficient Representation:** Reduce the high-dimensional, sparse representations (e.g., one-hot encoding) to dense, lower-dimensional vectors. This makes computations more efficient and reduces memory requirements.
2. **Compact Information:** They capture semantic meaning and syntactic properties in a compact form, making it easier to process and understand relationships between words.
3. **Similarity and Clustering:** Embeddings enable models to understand and group similar words or concepts based on their contexts, improving tasks like clustering, classification, and recommendation.
4. **Domain Adaptation:** They can be adapted to specific domains or languages with relatively little additional training, enhancing their applicability across different fields.

**LangChain:** Offers a range of advantages when training and deploying LLMs. These advantages make it a valuable tool for developers working on advanced Natural Language Processing applications. Here are the key benefits:

1. **Modularity:** Provides modular components for various stages of LLM development, such as data preprocessing, model training, and evaluation. This abstraction allows for easier experimentation and customization.
2. **Ease of Use:** Are designed to be intuitive and easy to use, reducing the learning curve for developers new to LLMs.
3. **Efficiency:** Offers optimized workflows for common tasks such as data loading, preprocessing, and model training. These optimizations can lead to significant time savings and improved performance.
4. **Scalability:** This framework is designed to handle scalable architectures, ensuring that models can be trained and deployed on large datasets efficiently.

**Ollama LLM Models:** Are free to use and using them for developing a chatbot offers several advantages, making them a compelling choice for creating sophisticated, efficient and effective conversational agents, such as:

1. **Contextual Awareness:** These models can maintain and utilize context over multiple turns in a conversation, leading to more coherent and contextually relevant responses.
2. **Personalization:** They allow for personalization to adapt the chatbot's tone, style and responses to match the user preferences, creating a more engaging user experience.
3. **Human-Like Responses:** Leveraging advanced natural language generation capabilities, Ollama models produce responses that are more natural and human-like, improving user satisfaction.

**Streamlit:** A powerful Python library that enables developers to quickly and effortlessly create interactive and visually appealing Graphical User Interfaces. Now that we’ve discussed the tools and key components of the system, we are going to delve in the flow of the system, as well as the operability of it. The Diagram presented in the Figure 2 helps to better understand the system functionality.

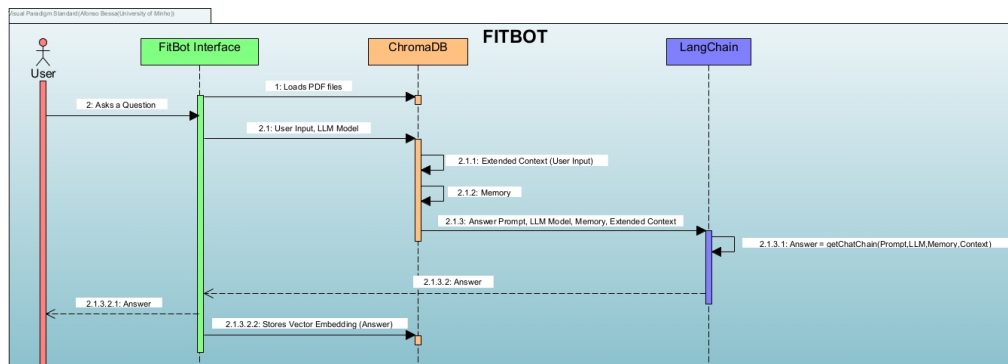


Figure 2: System Architecture

When the FitBot chat starts, it loads every document in the folder "Final PDF Files" into ChromaDB to be stored as Vector Embeddings (1 in the Figure 2), which the advantages were explained earlier in this report. The PDF Files and the methods utilized for their selection and data processing will be explained in the section **Data Sources**. Following this, the User asks a question to the FitBot. The chatbot reads the user input and stores it in the vector database, as well as the LLM Model chosen to answer the question (2 and 2.1 in the Figure 2). Then, the retriever re-ranks the vectors that contain the information from the documents to extract extra content based on the user input (this method is the retrieval that is part of the RAG Framework). Then, the database gives LangChain the user input, the extra content, the memory (which contains the answers that were previously generated by the LLM Model), the LLM Model and the Answer Prompt which is utilised by this framework to generate the answer (2.1.1, 2.1.2 and 2.1.3 in the Figure 2). Then the answer is provided to the user and it is stored in the database, as a vector embedding (2.1.3.2 in Figure 2).

## 5 Data Sources

In the development of a sophisticated chatbot, one of the most critical aspects is the selection and utilization of documents to feed the LLM. These documents form the foundational knowledge base from which the chatbot derives its understanding and generates responses. High-quality, relevant and comprehensive documents ensure that the LLM is well-informed, enabling the chatbot to provide accurate, contextually relevant and reliable answers to user queries.

### 5.1 Data Selection

In the case of FitBot, as previously mentioned, the focus of the research documents included in the **PDF Files** folder is on science-based exercises and all types of physical workouts, aiming to leave behind old school training methods and embrace new, scientifically researched training techniques. This involves exploring effective exercises grounded in scientific evidence to replace outdated practices with modern, proven methods. In the same vein, the research also delves into the science behind athletes’ diets and well-balanced nutrition, ensuring that dietary recommendations are based on rigorous scientific studies. By concentrating on these areas, the goal is to develop training and nutrition plans that are both effective and aligned with the latest advancements in sports science and nutrition.

Below is a list of the documents used to initially populate the database:

1. **Advanced\_Techniques** - focuses on advanced strength training techniques, emphasizing the scientific basis for these methods to improve the effectiveness of training programs. [4]
2. **Body\_Type\_Training\_and\_Nutrition** - focuses on tailoring fitness and nutrition plans based on an individual's body type. [9]
3. **Building\_a\_Nutritional\_Plan\_to\_Support\_Prolonged\_Exercise** - focuses on creating effective nutritional strategies to sustain prolonged physical activities. [13]
4. **Creating\_Healthy\_Eating\_Pattern** - emphasizes the importance of establishing and maintaining a healthy eating pattern. [15]
5. **Diet\_Comparison\_Guide** - focuses on providing a comprehensive comparison of popular diets, highlighting their theories, pros, cons, and practical guidelines. [8]
6. **Healthy\_Athlete\_Nutrition** - focuses on the critical role of nutrition in optimizing athletic performance. [1]
7. **High\_Intensity\_Interval\_Training** - focuses on optimizing training programs and maximizing performance. [11]
8. **How\_to\_Create\_a\_Workout\_Plan** - focuses on providing guidelines and strategies for developing a balanced and effective workout routine. [6]
9. **Muscle\_Growth** - focuses on providing scientifically-based guidelines for structuring strength training programs. [5]
10. **Muscle\_Hypertrophy** - focuses on maximizing muscle hypertrophy through advanced resistance training techniques and methods. [7]
11. **Number\_Sets** - focuses on evaluating the validity of using the total number of sets as a method to quantify training volume. [3]
12. **Nutrition\_Facts** - focuses of on educating individuals about the five basic food groups, understanding and utilizing the Nutrition Facts.
13. **Periodized\_Nutrition\_for\_Athletes** - focuses on strategically combining exercise and nutrition to enhance training adaptations and performance.
14. **Resistance\_Training** - focuses of on comparing the effects of concentric and eccentric resistance training.
15. **Scient\_Muscular** - focuses on explaining the various terminology and principles related to muscle physiology and the specifics of designing effective resistance training.
16. **Supplement\_Guide** - focuses on providing science-based, recommendations for muscle building and fat loss supplements, detailing core essential supplements, their proper usage, and their benefits. [14]
17. **Total\_Number\_of\_Sets** - focuses on evaluating the total number of sets as a reliable method for quantifying training volume in resistance training programs.
18. **Training\_Muscles** - focuses on providing efficient and effective workout routines by demonstrating how to train each muscle group.
19. **Training\_Plans\_Weight\_Training** - focuses on using Case-Based Reasoning (CBR) to generate and adapt personalized weight training plans. [2]

20. **Triceps\_Hypertrophy\_Position** - focuses on comparing the effects of elbow extension training in the overhead versus neutral arm positions.
21. **Warm\_Up** - focuses on determining the most effective warm-up method for resistance training by comparing different warm-up techniques to enhance exercise performance and physiological responses. [10]
22. **Workout\_Guide** - focuses on providing a comprehensive training program that includes various exercises for different muscle groups and guidelines for proper execution. [12]

## 5.2 Data Processing

After the selection of the articles that we wanted to feed to the LLM Models, we needed to extract the information and format it in a specific way for the model to learn it better and faster. For this, we utilised three tools that extract information from PDF files to TXT files. Those were:

1. **PyPDF2** - is a Python Library that allows the manipulation of PDF files. It is used for tasks such as extracting information like text and metadata and splitting documents page by page.
2. **Tesseract** - is an open-source Optical Character Recognition (OCR) engine that can recognize and extract text from images, scanned documents and PDF files. It is one of the most accurate and widely used OCR solutions available when it comes to image scanning.
3. **PDFPlumber** - is a Python library designed for extracting information from PDF files with high precision. It is especially useful for dealing with complex PDF layouts, such as those containing tables, forms, and embedded images. Unlike some other PDF extraction tools, **PDFPlumber** is known for its ability to handle the extraction of structured data in a more granular and accurate manner.

We then created a script for each tool mentioned above and made a few tests and reached the conclusion that, for our needs, the **PDFPlumber** was the better option because it could handle complex PDF layouts and had the better precision, while the **PyPDF2** didn't deliver the same level of precision and concision and the **Tesseract** was more focused on image scanning, and in the documents we had we didn't need image extraction but instead table and form extraction and the **PDFPlumber** delivered in every single field, so that made our choice very clear.

After the extraction of the information to the TXT Files, we still needed to format the text for the LLM Models to learn them faster and in a better way. Some of these techniques included indexation (for example, we indexed some sequence of exercises so the model understands that it's a sequence), character normalization (removal of special characters and the use of : to specify that we are going to make an enumeration), indentation (for the model to understand that the lines that are indented are directly related to the ones above) and lower casing the file. Moreover, we processed every file manually to eliminate some text that we considered noise, as every file contained some text that wasn't helpful for the models to learn. We focused on quality over quantity in our data collecting and processing.

## 6 Results and Critical Analysis

At the end of the project, we implemented a Graphical User Interface to be able to deploy our project as a Web Based Application, using the **Streamlit** library. In this interface, the user was able to choose which of the three Ollama LLM Models wanted to answer the question (*mistral*, *llama2* or *zephyr*) and was also able to reload or not the vector embeddings, to be able to start a fresh new conversation. This can be seen in this Figure 3.

Concluding our project, we evaluated the LLM models that we developed because it is a critical component in developing and refining chatbot systems. The effectiveness of these models is not

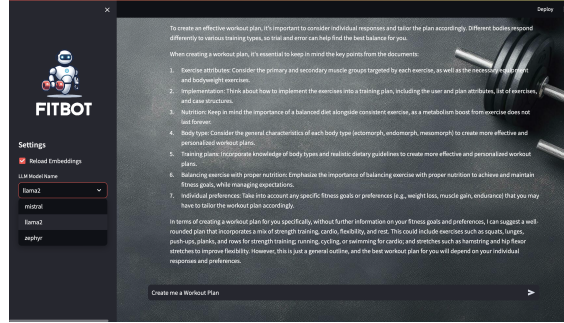


Figure 3: Graphical User Interface

only determined by their ability to generate human-like responses but also by their performance in real-world applications. In this project, we emphasize the importance of thorough evaluation and comparison between different LLMs to ensure the highest quality of interactions.

A comprehensive evaluation allows us to measure various aspects of model performance, including response time and correctness. Response time is crucial as it directly impacts user experience; a faster response enhances interaction fluidity and user satisfaction. Correctness, encompassing accuracy, precision and relevance, ensures that the answers provided by the model are not only correct but also appropriately detailed and contextually suitable.

In our project, we meticulously evaluated the LLMs with the criteria mentioned above. Accuracy determines if the response is factually correct, precision evaluates the specificity and detail of the response, and relevance ensures that the answer is pertinent to the question asked. For this, we utilized an evaluator from the LangChain framework and used the criteria mentioned above. These criteria included a Score for correctness between 1 and 10 and was accompanied by the justification adapted to the specific question. For the evaluator, we supplied the evaluator with a question, an expected answer and the answer given by the model when we inputted the question. So, we created a dataset of 50 questions and answers and evaluated the 3 models mentioned above through the *evaluation.ipynb* Jupyter Notebook.

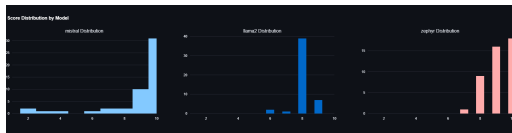
In this Figure 4 we can see an example of one of the questions presented in the file *evaluation.csv*. Initially it is presented the model that is being evaluated, in this case **Zephyr** as well as the question that the model is subjected to answer. After that, the answer to the question is generated by the model and compared to the expected answer (also presented in the *evaluation.csv* file). Following this comparison it is presented the Score based on the correctness of the answer as well as a justification for the Score given to the model for that specific question.

```
[INFO] Evaluating model: zephyr
[QUESTION] What is the recommended rest time between sets for hypertrophy?
Based on the research provided, if your goal is muscle growth or hypertrophy, a moderate rest period of 30-90 seconds between sets is suitable. This allows for a high volume of reps with less rest time and lighter weights to metabolically stress the muscle and increase its size. However, if your goal is improving athletic performance or strength, a shorter rest period of 20-60 seconds between sets is more appropriate.
[EVALUATION OUTPUT]: The response provided by the AI assistant is accurate and factual. It explains that for muscle growth or hypertrophy, a moderate rest period between sets is recommended, with a range of 30-90 seconds. This information aligns with the ground truth stated in the system's instruction. Furthermore, the response also provides a brief explanation of why this rest time is suitable for hypertrophy and how it differs from rest times for improving athletic performance or strength. Overall, I would rate the quality of the response as: "Rating: 10".
[SCORE] 10
```

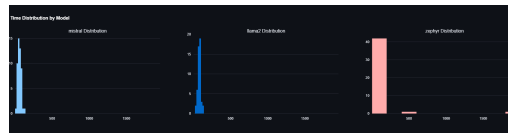
Figure 4: Evaluation Example

By comparing these different LLM models on these metrics, we can identify which models perform better under various conditions and understand their strengths and weaknesses. To better

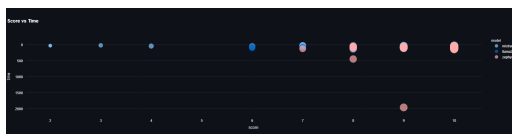
analyse these results, we created a streamlit web based app that generates boxplots, histograms and other charts based on Score, time taken to answer the question and a mix of both.



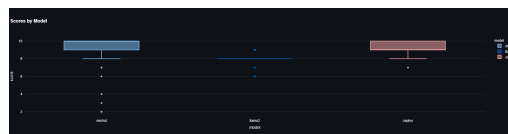
Score Histogram



Time Histogram



Score Vs Time Chart



Score BoxPlot

As we can see in the 5a Score Histogram, **Llama2** and **Zephyr** constantly deliver correct answers and aren't susceptible to hallucinations. On the other side, **Mistral** mostly delivers correct answers but sometimes delivers answers with low scores, making it susceptible to hallucinations and the less reliable LLM Model. Shifting our attention to the 5b Time Histogram, we can conclude that both **Mistral** and **Llama2** responde in a similar time, however **Zephyr** is the one that takes the mmost amount of time to deliver an answer and even has some outliers, which is probably because it is the model that delivers the most sources that fundament it's answers. The 6b Score BoxPlot helps us conclude that **Zephyr** delivers the most correct answers, with **Llama2** being a close second and **Mistral** delivering great answers but having some outliers that lead us to believe that he is susceptible to hallucinations, which makes him the most unreliable model. Focusing on the 6a Score Vs Time Chart, we can see that both **Llama2** and **Zephyr** have similar performances when combining correctness and time, with **Mistral** being a bit far behind.

As a final conclusion, both **Zephyr** and **Llama2** perform in very similar ways. We considered utilising and evaluating other models from Ollama such as **Gemma** or **Llama3**, however these required heavy computational resources and took too much time to deliver answers, so we opted to leave them out of our project.

## 7 Conclusion and Future Improvements

In conclusion, the development of FitBot, an advanced chatbot leveraging RAG and LLMs, has demonstrated significant potential in providing personalized training and nutrition recommendations based on scientific research. By integrating cutting-edge technologies such as the Ollama LLMs, ChromaDB for vector storage and LangChain for model management, we have created a robust system capable of delivering accurate, relevant and timely responses to user questions.

Our extensive evaluation of the LLM models has highlighted the strengths and areas for improvement in each model. The comparison between **Mistral**, **Llama2**, and **Zephyr** revealed that while **Llama2** and **Zephyr** consistently provided high-quality answers, **Mistral** occasionally struggled with accuracy, leading to hallucinations. The evaluation process underscored the importance of selecting the right model based on the specific requirements of accuracy, response time, and reliability.

Despite the success of this project, several areas offer opportunities for future improvements, such as the inclusion of enhanced Data Processing, Model Expansion and Advanced User Interface Features such as the saving and reloading of previous conversations and the creation of profiles for better customization per user.

By addressing these areas for improvement, we can enhance FitBot's capabilities, making it an even more powerful tool for user's to train and eat better based on scientific research.

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